

URAKAWA, Japan

WMO: 474260

Lat: 42.1625N

Lon: 142.7769E

Elev: 38

StdP: 100.86

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-7.9	-17.1	0.8	-7.8	-15.5	1.0	-6.8	18.0	-0.8	15.6	-1.3	4.6	310	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	24.8	22.6	23.8	22.0	22.9	21.4	23.2	24.3	22.4	23.5	21.7	22.7	4.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	17.6	24.0	22.1	16.8	23.3	21.3	16.0	22.5	69.3	24.4	66.5	23.5	63.6	22.7	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.3	12.1	10.2	-11.1	26.9	1.8	1.5	-12.4	28.0	-13.4	28.9	-14.4	29.8	-15.7	30.9		
			-12.3	24.1	1.7	1.1	-13.5	24.9	-14.6	25.5	-15.5	26.1	-16.8	26.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-2.6	-2.3	0.8	5.2	9.9	13.6	17.9	20.2	17.9	12.3	6.1	0.2	
	DBStd	8.32	2.64	2.67	2.61	2.59	2.59	2.54	2.44	2.25	2.65	2.95	3.73	3.25	
	HDD10.0	1643	389	345	284	146	34	3	0	0	0	13	126	304	
	HDD18.3	3767	648	578	543	393	262	142	36	8	39	188	368	562	
	CDD10.0	1035	0	0	0	2	31	112	246	316	238	83	8	0	
	CDD18.3	118	0	0	0	0	0	1	24	66	27	0	0	0	
	CDH23.3	142	0	0	0	0	0	0	19	87	36	0	0	0	
	CDH26.7	3	0	0	0	0	0	0	1	2	1	0	0	0	
Wind	WSAvg	4.3	5.2	5.1	5.0	4.2	3.7	3.1	2.9	3.0	3.8	4.7	5.3	5.4	
Precipitation	PrecAvg	1134	41	32	53	85	117	109	140	156	140	107	89	63	
	PrecMax	1502	87	87	145	225	229	273	378	430	293	238	214	148	
	PrecMin	646	6	2	8	10	19	25	19	29	44	23	10	14	
	PrecStd	186	21	20	29	41	50	60	79	89	58	50	41	31	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	6.4	9.5	15.3	18.4	21.2	25.1	26.6	26.2	20.6	16.1	11.0	
		MCWB	4.2	5.1	6.9	10.1	14.0	17.9	22.7	23.7	23.4	17.6	13.9	9.5	
	2%	DB	4.0	4.2	7.5	12.3	16.6	19.5	23.6	25.3	24.2	19.2	14.6	9.0	
		MCWB	2.2	2.5	5.4	9.4	13.1	17.1	21.9	23.0	22.1	16.6	12.6	7.3	
	5%	DB	2.7	3.0	6.1	10.7	15.2	18.4	22.5	24.3	23.2	18.1	13.3	7.1	
		MCWB	0.8	1.2	4.2	8.3	12.5	16.4	21.0	22.3	21.2	15.4	11.2	5.1	
	10%	DB	1.6	2.0	4.9	9.3	14.0	17.4	21.4	23.4	22.1	17.1	12.0	5.3	
		MCWB	-0.4	0.2	3.0	7.2	11.7	15.8	20.2	21.8	20.2	14.5	9.8	3.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	5.3	7.8	11.5	15.5	18.8	23.2	24.2	24.1	18.9	14.8	9.8
			MCDB	5.5	5.9	9.0	13.3	17.1	20.0	24.4	25.8	25.2	19.7	15.7	10.7
		2%	WB	2.7	3.0	5.9	9.9	14.1	17.7	22.3	23.4	22.8	17.1	13.0	7.6
			MCDB	3.7	3.9	7.2	11.7	15.6	18.9	23.3	24.7	23.8	18.5	14.2	8.8
5%		WB	1.1	1.6	4.4	8.7	13.0	16.8	21.3	22.8	21.6	16.0	11.6	5.3	
		MCDB	2.4	2.7	5.8	10.4	14.6	18.0	22.2	23.8	22.8	17.6	13.1	7.0	
10%		WB	-0.2	0.4	3.3	7.5	12.1	16.0	20.4	22.2	20.6	14.9	10.0	3.2	
		MCDB	1.4	1.8	4.7	9.0	13.7	17.2	21.3	23.1	21.9	16.7	11.8	5.0	
Mean Daily Temperature Range		5% DB	MDBR	5.9	6.0	5.9	6.0	5.9	5.2	4.5	4.7	6.1	7.0	6.6	5.9
			MCDBR	6.4	6.2	6.9	7.7	7.6	6.5	5.3	5.3	6.1	7.1	7.3	7.6
			MCWBR	6.0	5.9	5.9	5.8	5.2	4.5	3.7	3.5	4.4	5.8	7.1	7.4
		5% WB	MCDBR	6.4	6.4	6.8	7.4	6.8	5.9	5.1	4.8	5.3	6.6	7.6	7.7
	MCWBR		6.6	6.5	6.2	5.8	5.2	4.4	3.7	3.4	4.2	6.0	7.9	8.0	
	Clear-Sky Solar Irradiance		taub	0.339	0.385	0.461	0.515	0.532	0.503	0.496	0.474	0.416	0.400	0.382	0.355
taud		2.172	2.031	1.891	1.838	1.890	2.052	2.148	2.225	2.343	2.249	2.191	2.155		
Ebn at Noon		774	788	768	760	760	784	786	791	814	772	715	711		
Edh at Noon		99	135	175	199	194	165	149	134	111	108	98	94		
All-Sky Solar Radiation	RadAvg	1.78	2.64	3.78	4.72	5.04	4.97	4.50	4.17	3.70	2.87	1.81	1.42		
	RadStd	0.10	0.21	0.27	0.44	0.46	0.37	0.43	0.34	0.30	0.17	0.14	0.12		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page